Storyline

Chapter 8: Conservation of Energy



Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

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Analysis Model: Nonisolated System (Energy)

Nonisolated system: energy crosses the boundary of the system during some time interval due to interaction with environment

Energy Transfer Mechanisms

Energy is transferred to the block by *work*.



a

Energy leaves the radio from the speaker by *mechanical waves*.



b

Energy transfers to the handle of the spoon by *heat*.



c

Energy enters the automobile gas tank by *matter transfer*.



d

Energy enters the hair dryer by *electrical transmission*.



e

Energy leaves the light-bulb by *electromagnetic radiation*.



f

Analysis Model: Nonisolated System (Energy)

Energy is always *conserved*.

$$\Delta E_{\text{system}} = \sum T$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}}$$

If only change in system is speed of one or more components:

Work-kinetic energy theorem: $\Delta K = W$

Quick Quiz 8.1 Part I

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *block,* this system is

(a) isolated

(b) nonisolated

(c) impossible to determine

Quick Quiz 8.1 Part I

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *block,* this system is

(a) isolated

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Quick Quiz 8.1 Part II

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *surface*, this system is

(a) isolated

(b) nonisolated

(c) impossible to determine

Quick Quiz 8.1 Part II

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *surface,* this system is

(a) isolated

(b) nonisolated

(c) impossible to determine

Quick Quiz 8.1 Part III

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *block and the surface,* this system is

(a) isolated

(b) nonisolated

(c) impossible to determine

Quick Quiz 8.1 Part III

Consider a block sliding over a horizontal surface with friction. Ignore any sound the sliding might make.

If the system is the *block and the surface,* this system is

(a) isolated

(b) nonisolated

(c) impossible to determine

Analysis Model: Nonisolated System (Energy)

$$\Delta E_{\text{system}} = \sum T$$

Microscopic Work

Mass Transfer

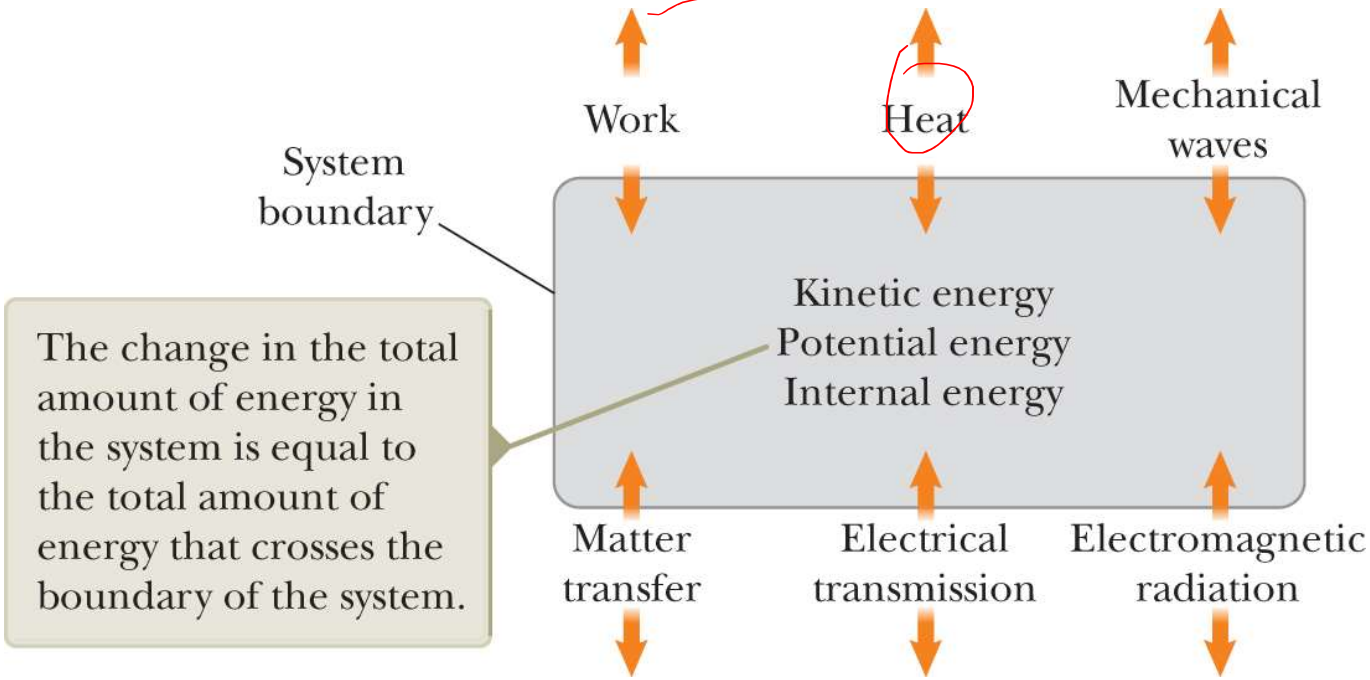
Energy Transport

Energy Redistribution

Internal Energy

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}}$$

Energy transfer



Conceptual Example 8.1: Reducing the Conservation of Energy Equation in Specific Situations

When using the energy conservation equation to solve a problem, the following steps should be remembered:

(1) define the system;

(2) identify the beginning and end of the time interval of interest;

(3) identify initial and final configurations of the system (positions of objects in gravitational situations, extensions of springs, etc.) and assign appropriate reference values of potential energy;

(4) write the energy conservation equation, eliminating or setting terms equal to zero that do not apply in the situation.

Conceptual Example 8.1: Reducing the Conservation of Energy Equation in Specific Situations

Consider the following examples. For each example, the system is provided and the time interval is from before the device is turned on until it has been operating for a few moments.

(A) Your television set.

$$\cancel{\Delta K} + \cancel{\Delta U} + \Delta E_{\text{int}} = \cancel{W} + Q + T_{\text{MW}} + \cancel{T_{\text{MT}}} + T_{\text{ET}} + T_{\text{ER}} \rightarrow$$

$$\Delta E_{\text{int}} = Q + T_{\text{MW}} + T_{\text{ET}} + T_{\text{ER}}$$

Conceptual Example 8.1: Reducing the Conservation of Energy Equation in Specific Situations

(B) Your gasoline-powered lawn mower. The time interval includes the process of filling the tank with gasoline.

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}} \rightarrow$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}}$$

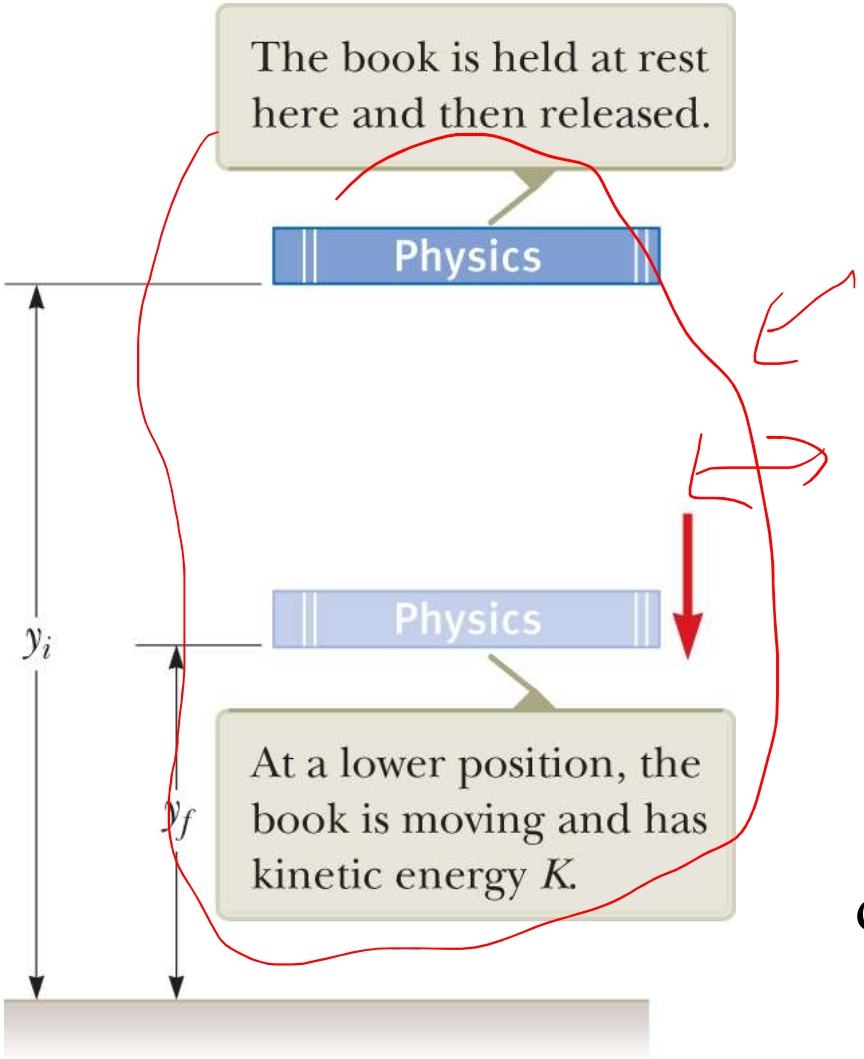
Conceptual Example 8.1: Reducing the Conservation of Energy Equation in Specific Situations

(C) A cup of tea being warmed in a microwave oven.

$$\cancel{\Delta K} + \cancel{\Delta U} + \Delta E_{\text{int}} = \cancel{W} + Q + \cancel{T_{\text{MW}}} + \cancel{T_{\text{MT}}} + \cancel{T_{\text{ET}}} + T_{\text{ER}} \rightarrow$$

$$\Delta E_{\text{int}} = Q + T_{\text{ER}}$$

Analysis Model: Isolated System (Energy)



$$\Delta K + \Delta U_g = 0$$

$$\Delta K + \Delta U = 0$$

Transformation mechanism: work done on book within system by the gravitational force

Conservation of Mechanical Energy

$$E_{\text{mech}} \equiv K + U$$

$$\Delta E_{\text{mech}} = 0$$

$$\left(K_f - K_i\right) + \left(U_f - U_i\right) = 0$$

$$K_f + U_f = K_i + U_i$$

→ For N-particle

$$\frac{1}{2}mv_f^{\,2} + mgy_f = \frac{1}{2}mv_i^{\,2} + mgy_i$$

$$E_{\text{mech}} + E_{\text{internal}} = E_{\text{total},i} = E_{\text{total},f} \qquad \Delta E_{\text{system}} = 0$$

Quick Quiz 8.2

A rock of mass m is dropped to the ground from a height h. A second rock, with mass $2m$, is dropped from the same height. When the second rock strikes the ground, what is its kinetic energy?

(a) twice that of the first rock

(b) four times that of the first rock

(c) the same as that of the first rock

(d) half as much as that of the first rock

(e) impossible to determine

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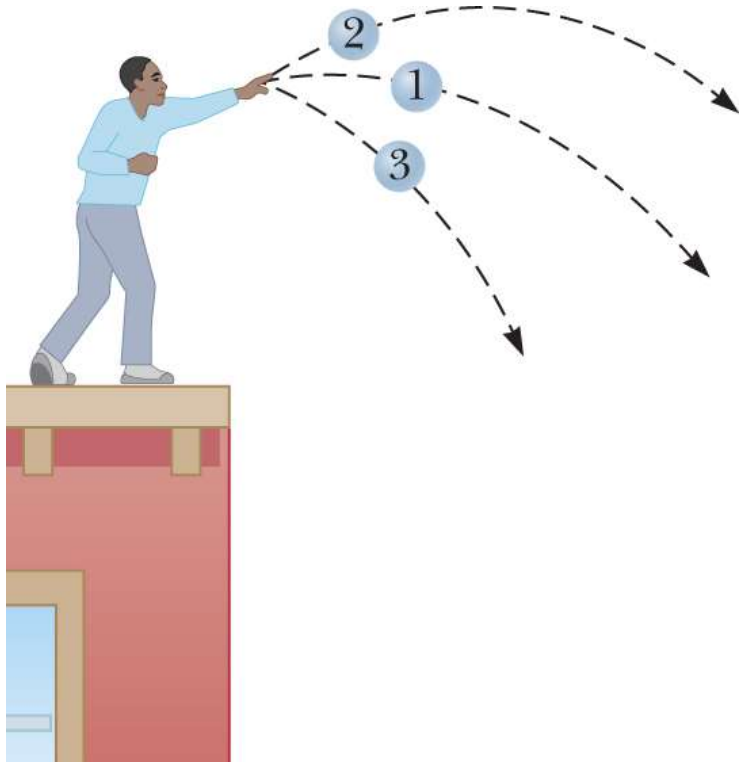
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(d) half as much as that of the first rock

(e) impossible to determine

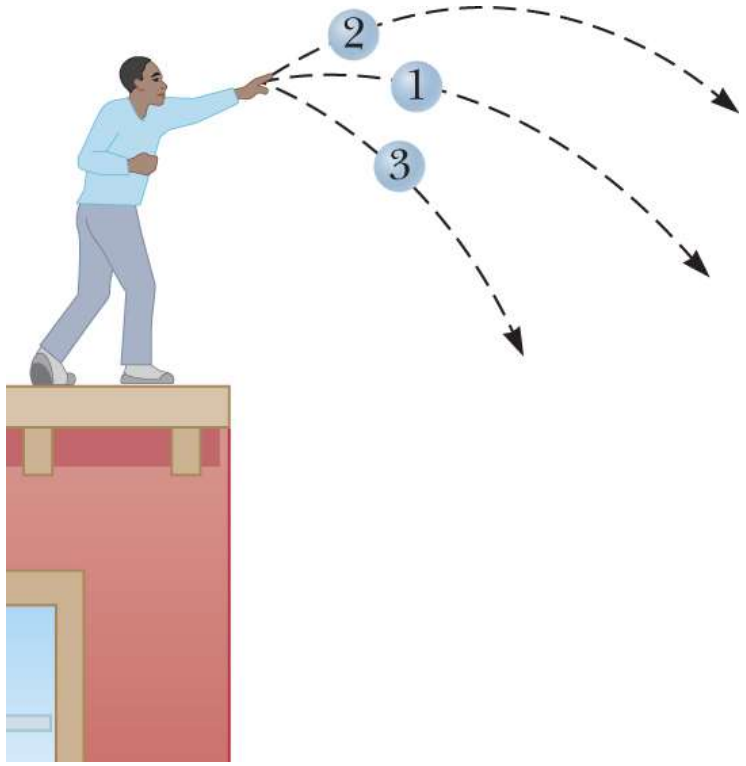
Quick Quiz 8.3

Three identical balls are thrown from the top of a building, all with the same initial speed. As shown in the figure, the first is thrown horizontally, the second at some angle above the horizontal, and the third at some angle below the horizontal. Neglecting air resistance, rank the speeds of the balls at the instant each hits the ground.



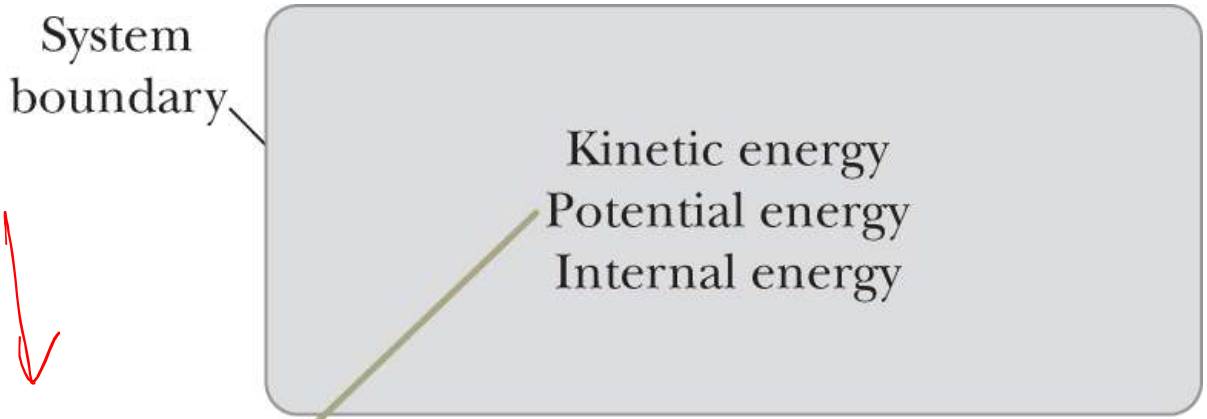
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$$v_1 = v_2 = v_3$$

Analysis Model: Isolated System (Energy)



The total amount of energy in the system is constant. Energy transforms among the three possible types.

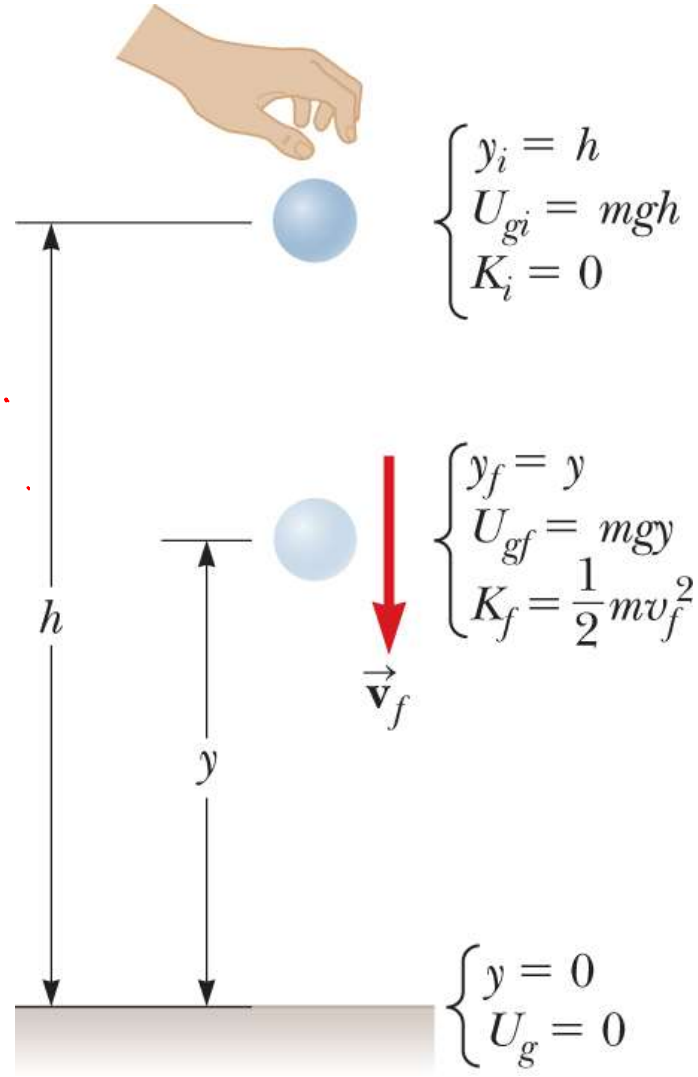
$$\Delta E_{\text{system}} = 0$$

If no nonconservative forces act: $\Delta E_{\text{mech}} = 0$

Example 8.2:
Ball in Free Fall

A ball of mass m is dropped from a height h above the ground as shown in the figure.

(A) Neglecting air resistance, determine the speed of the ball when it is at a height y above the ground. Choose the system as the ball and the Earth.



Example 8.2:
Ball in Free Fall

$$\Delta K + \Delta U_g = 0$$

$$\left(\frac{1}{2}mv_f^{\,2} - 0\right) + \left(mgy - mgh\right) = 0$$

$$v_f^{\,2} = 2g\left(h - y\right) \Rightarrow v_f = \sqrt{2g\left(h - y\right)}$$

Example 8.2:
Ball in Free Fall

(B) Find the speed of the ball again at height y by choosing the ball as the system.

$$\Delta K = W$$

$$\left(\frac{1}{2}mv_f^{\,2} - 0\right) = \vec{\mathbf{F}}_g \cdot \Delta\vec{\mathbf{r}} = -mg\hat{\mathbf{j}} \cdot \Delta y\hat{\mathbf{j}}$$

$$= mg\Delta y = -mg\left(y - h\right) = mg\left(h - y\right)$$

$$v_f^{\,2} = 2g\left(h - y\right) \Rightarrow v_f = \sqrt{2g\left(h - y\right)}$$

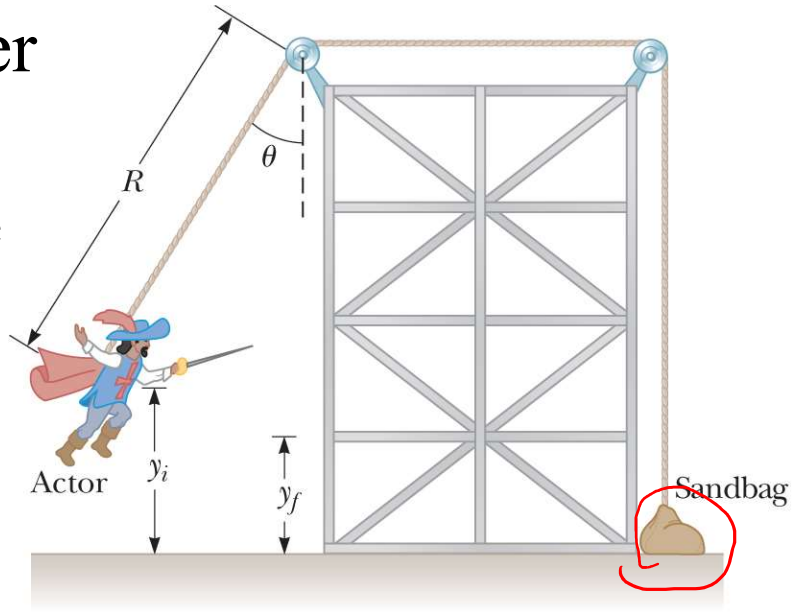
Example 8.2:
Ball in Free Fall

What if the ball were thrown downward from its highest position with a speed v_i? What would its speed be at height y?

$$v_f = \sqrt{v_i^2 + 2g(h-y)}$$

Example 8.3:
A Grand Entrance

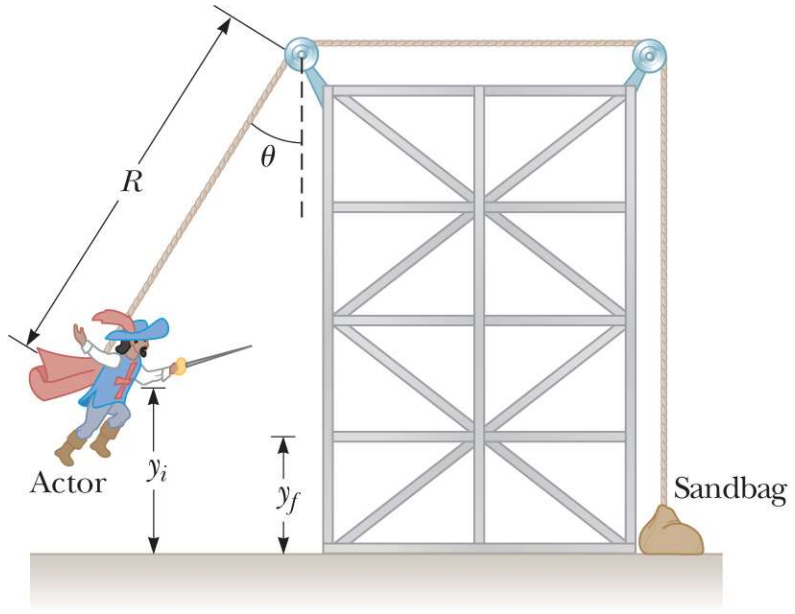
You are part of the stage crew for a theatrical company and are designing an apparatus to support an actor of mass 65.0 kg who is to "fly" down to the stage during the performance of a play. You attach the actor's harness to a 130-kg sandbag by means of a lightweight steel cable running smoothly over two frictionless pulleys as in the figure. You need 3.00 m of cable between the harness and the nearest pulley so that the pulley can be hidden behind a curtain.



Example 8.3:
A Grand Entrance

For the apparatus to work successfully, the sandbag must never lift above the floor as the actor swings from above the stage to the floor. Let us call the initial angle that the actor's cable makes with the vertical θ.

What is the maximum value θ can have before the sandbag lifts off the floor?



Example 8.3:
A Grand Entrance

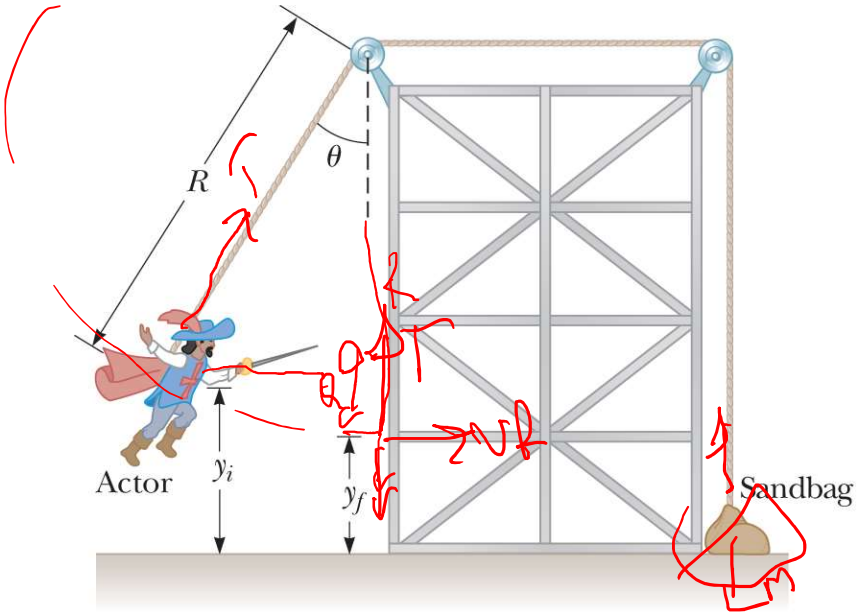
$$\Delta K + \Delta U_g = 0$$

$$\left(\frac{1}{2} m_{\text{actor}} v_f^{\,2} - 0\right) + \left(m_{\text{actor}} g y_f - m_{\text{actor}} g y_i\right) = 0$$

$$v_f^{\,2} = 2g\left(y_f - y_i\right)$$

$$\begin{aligned} y_f - y_i &= R - R\cos\theta \\ &= R\left(1 - \cos\theta\right) \end{aligned}$$

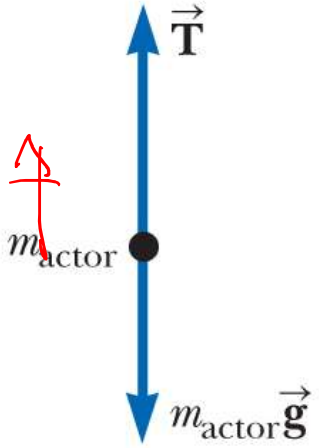
$$v_f^{\,2} = 2gR\left(1 - \cos\theta\right)$$



Example 8.3:
A Grand Entrance

$$\sum F_y = T - m_{\text{actor}} g = m_{\text{actor}} \frac{v_f^{\,2}}{R}$$

$$T = m_{\text{actor}} g + m_{\text{actor}} \frac{v_f^{\,2}}{R}$$



$$T = m_{\text{bag}} g \quad \text{and} \quad v_f^{\,2} = 2Rg\left(1 - \cos\theta\right)$$

$$\Rightarrow m_{\text{bag}} g = m_{\text{actor}} g + m_{\text{actor}} \frac{2gR\left(1 - \cos\theta\right)}{R}$$

$$\cos\theta = \frac{3m_{\text{actor}} - m_{\text{bag}}}{2m_{\text{actor}}} = \frac{3\left(65.0\ \text{kg}\right) - 130\ \text{kg}}{2\left(65.0\ \text{kg}\right)} = 0.500$$

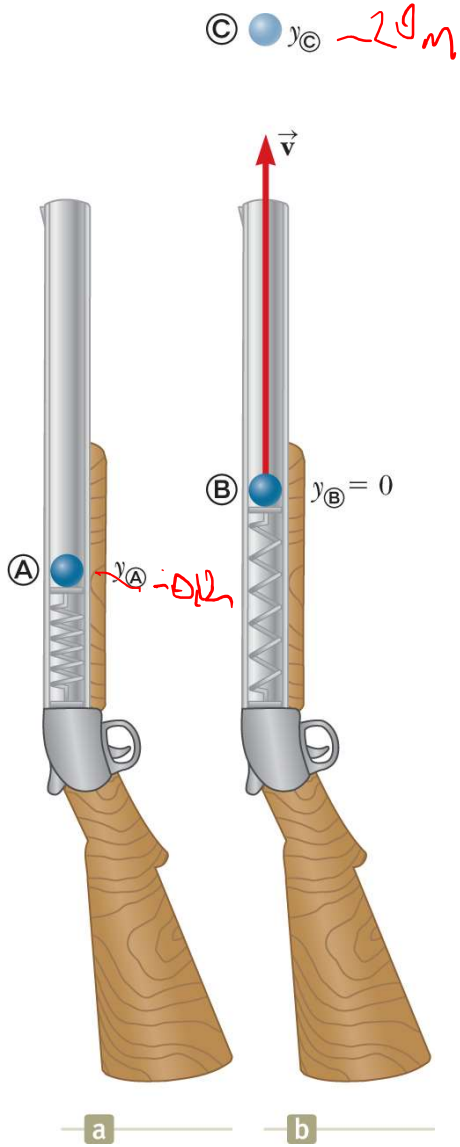
$$\theta = 60.0°$$



Example 8.4: The Spring-Loaded Popgun

The launching mechanism of a popgun consists of a trigger-released spring. The spring is compressed to a position y_A, and the trigger is fired. The projectile of mass m rises to a position y_C above the position at which it leaves the spring, indicated in the figure as position $y_B = 0$. Consider a firing of the gun for which $m = 35.0$ g, $y_A = -0.120$ m, and $y_C = 20.0$ m.

(A) Neglecting all resistive forces, determine the spring constant.



Example 8.3:
The Spring-Loaded Popgun

$$\Delta K + \Delta U_g + \Delta U_s = 0$$

$$(0-0) + (mgy_C - mgy_A) + \left(0 - \frac{1}{2}kx^2\right) = 0$$

$$k = \frac{2mg(y_C - y_A)}{x^2}$$

$$k = \frac{2(0.0350\ \text{kg})(9.80\ \text{m/s}^2)[20.0\ \text{m} - (-0.120\ \text{m})]}{(0.120\ \text{m})^2}$$

$$= \boxed{958\ \text{N/m}}$$

Example 8.3:
The Spring-Loaded Popgun

(B) Find the speed of the projectile as it moves through the equilibrium position B of the spring.

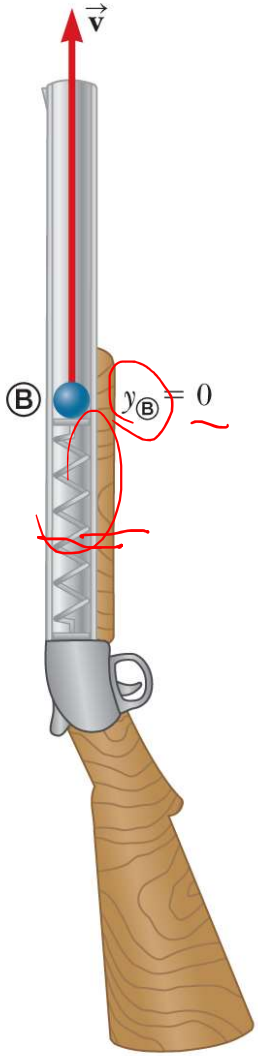
$$\Delta K + \Delta U_g + \Delta U_s = 0$$

$$\left(\frac{1}{2}mv_B^2 - 0\right) + \left(0 - mgy_A\right) + \left(0 - \frac{1}{2}kx^2\right) = 0$$

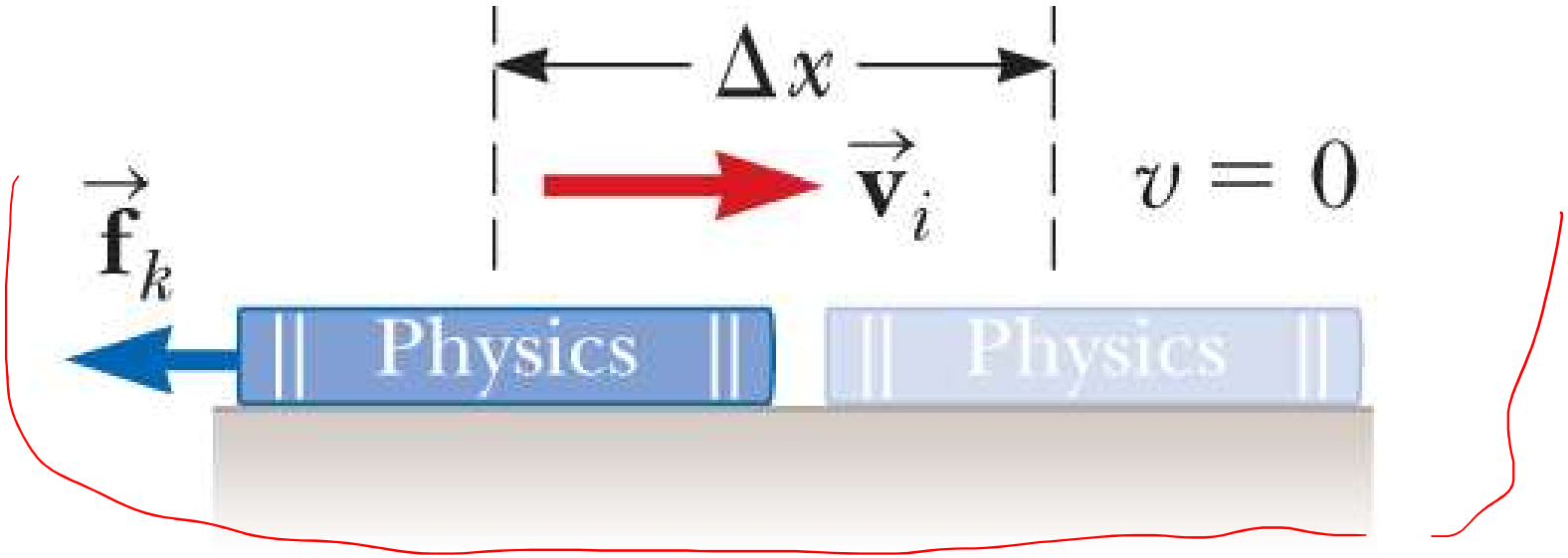
$$v_B = \sqrt{\frac{kx^2}{m} + 2gy_A}$$

$$v_B = \sqrt{\frac{(958\ \text{N/m})(0.120\ \text{m})^2}{(0.0350\ \text{kg})} + 2(9.80\ \text{m/s}^2)(-0.120\ \text{m})}$$

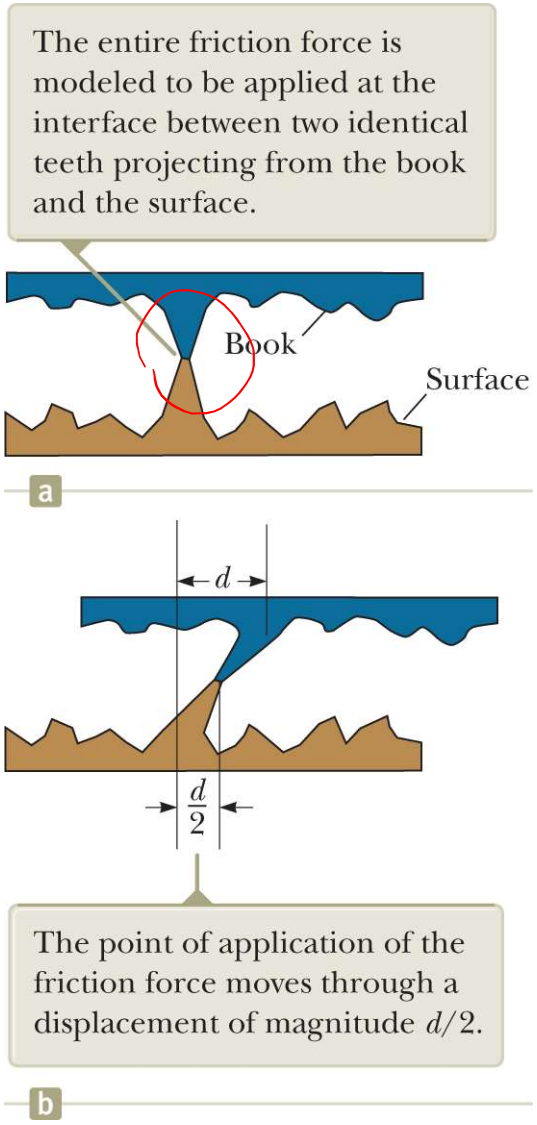
$$= \boxed{19.8\ \text{m/s}}$$



Situations Involving Kinetic Friction



Situations Involving Kinetic Friction



$$\sum W_{\text{other forces}} = \int \left(\sum \vec{\mathbf{F}}_{\text{other forces}} \right) \cdot d\vec{\mathbf{r}}$$

conservative forces

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}}$$

$$= \int \left(\sum \vec{\mathbf{F}}_{\text{other forces}} \right) \cdot d\vec{\mathbf{r}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}}$$

$$= \int \left(\sum \vec{\mathbf{F}}_{\text{other forces}} + \vec{\mathbf{f}}_k \right) \cdot d\vec{\mathbf{r}}$$

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \int \sum \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

Situations Involving Kinetic Friction

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \int \sum \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \int m\vec{\mathbf{a}} \cdot d\vec{\mathbf{r}}$$

$$= \int m \frac{d\vec{\mathbf{v}}}{dt} \cdot d\vec{\mathbf{r}} = \int_{t_i}^{t_f} m \frac{d\vec{\mathbf{v}}}{dt} \cdot \vec{\mathbf{v}} dt$$

$$\frac{d}{dt}\left(\vec{\mathbf{v}} \cdot \vec{\mathbf{v}}\right) = \frac{d\vec{\mathbf{v}}}{dt} \cdot \vec{\mathbf{v}} + \vec{\mathbf{v}} \cdot \frac{d\vec{\mathbf{v}}}{dt} = 2\frac{d\vec{\mathbf{v}}}{dt} \cdot \vec{\mathbf{v}}$$

$$\frac{d\vec{\mathbf{v}}}{dt} \cdot \vec{\mathbf{v}} = \frac{1}{2}\frac{d}{dt}\left(\vec{\mathbf{v}} \cdot \vec{\mathbf{v}}\right) = \frac{1}{2}\frac{dv^2}{dt}$$

Situations Involving Kinetic Friction

$$\frac{d\vec{\mathbf{v}}}{dt}\cdot\vec{\mathbf{v}} = \frac{1}{2}\frac{d}{dt}\left(\vec{\mathbf{v}}\cdot\vec{\mathbf{v}}\right) = \frac{1}{2}\frac{dv^2}{dt} \rightarrow$$

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \int m\vec{\mathbf{a}}\cdot d\vec{\mathbf{r}}$$

$$= \int m\frac{d\vec{\mathbf{v}}}{dt}\cdot d\vec{\mathbf{r}} = \int_{t_i}^{t_f} m\frac{d\vec{\mathbf{v}}}{dt}\cdot\vec{\mathbf{v}}dt$$

conservative non-cons

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \int_{t_i}^{t_f} m\left(\frac{1}{2}\frac{dv^2}{dt}\right)dt$$

$W_{net} = \Delta KE$

$$= \frac{1}{2}m\int_{v_i}^{v_f} d\left(v^2\right) = \frac{1}{2}mv_f{}^2 - \frac{1}{2}mv_i{}^2 = \Delta K$$

Situations Involving Kinetic Friction

$$\sum W_{\text{other forces}} + \int \vec{\mathbf{f}}_k \cdot d\vec{\mathbf{r}} = \Delta K$$

$$\sum W_{\text{other forces}} - \int f_k \, dr = \Delta K$$

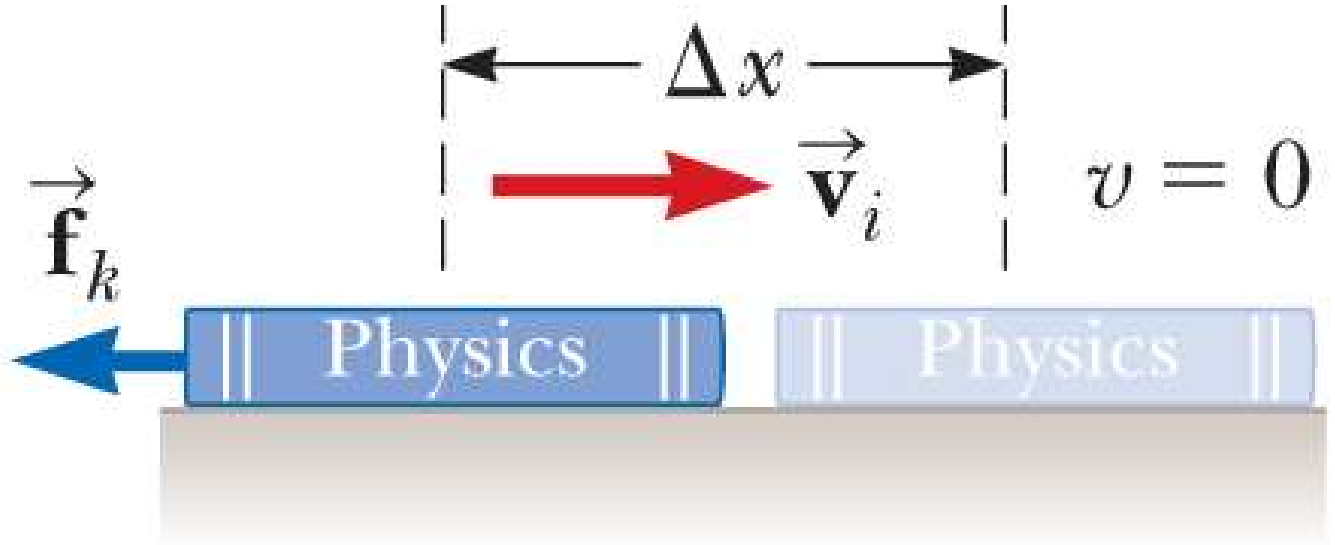
$$W - f_k d = \Delta K$$

Situations Involving Kinetic Friction

$$\Delta K + \Delta E_{\text{int}} = 0 \qquad -f_k d + \Delta E_{\text{int}} = 0 \qquad \Delta E_{\text{int}} = f_k d$$

Work done on system by forces other than friction:

$$W = \Delta K + \Delta E_{\text{int}}$$



Quick Quiz 8.4

You are traveling along a freeway at 65 mi/h. Your car has kinetic energy. You suddenly skid to a stop because of congestion in traffic. Where is the kinetic energy your car once had?

(a) It is all in internal energy in the road.

(b) It is all in internal energy in the tires.

(c) Some of it has transformed to internal energy and some of it transferred away by mechanical waves.

(d) It is all transferred away from your car by various mechanisms.

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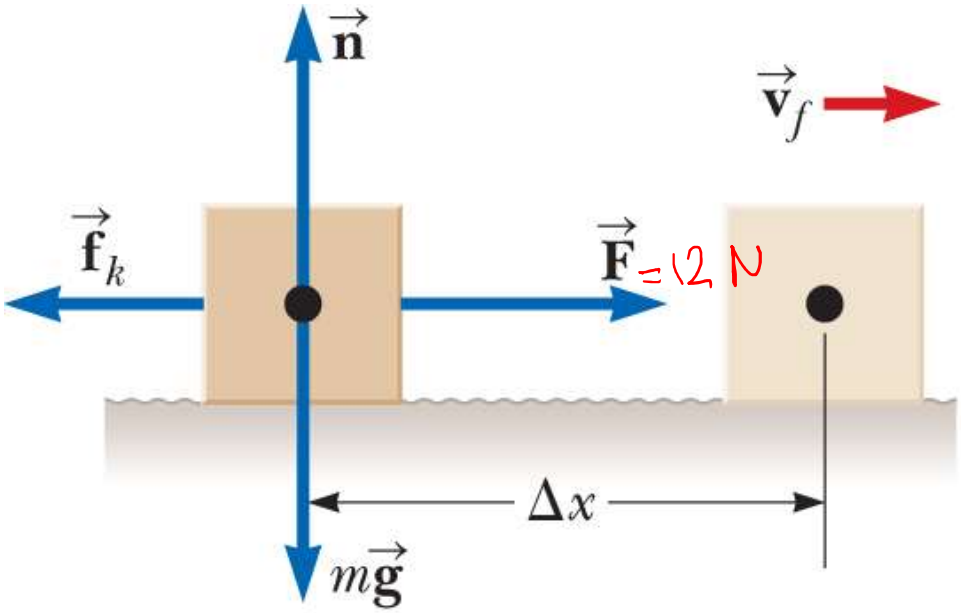
(c) Some of it has transformed to internal energy and some of it transferred away by mechanical waves.

(d) It is all transferred away from your car by various mechanisms.

Example 8.5:
A Block Pulled on a Rough Surface

A 6.0-kg block initially at rest is pulled to the right along a horizontal surface by a constant horizontal force of magnitude 12 N.

(A) Find the speed of the block after it has moved 3.0 m if the surfaces in contact have a coefficient of kinetic friction of 0.15.



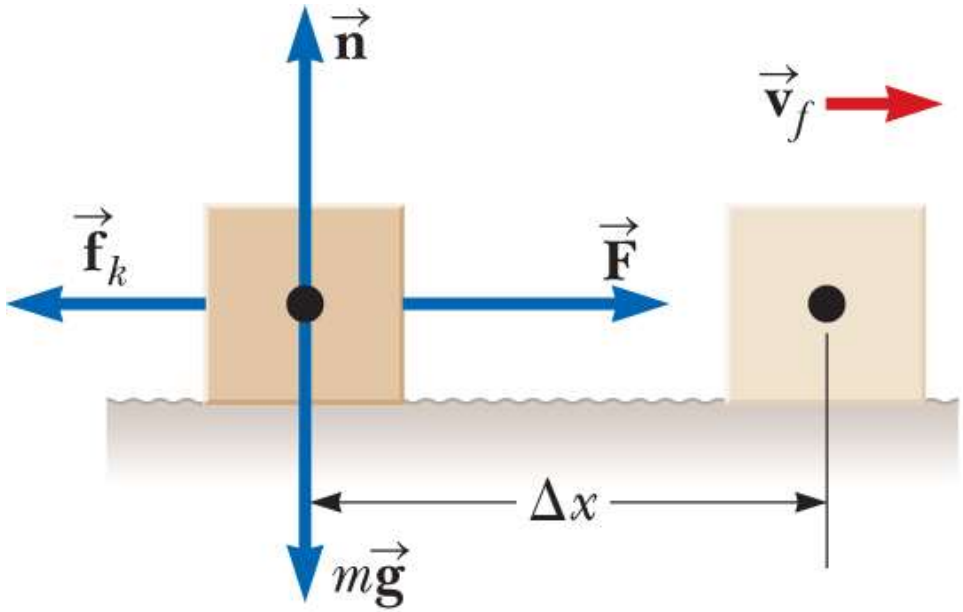
Example 8.5:
A Block Pulled on a Rough Surface

Box + surface is the system

$$\Delta K + \Delta E = W \qquad W = F\Delta x = Fd$$

$$\sum F_y = 0 \Rightarrow n - mg = 0 \Rightarrow n = mg$$

$$f_k = \mu_k n = \mu_k mg$$



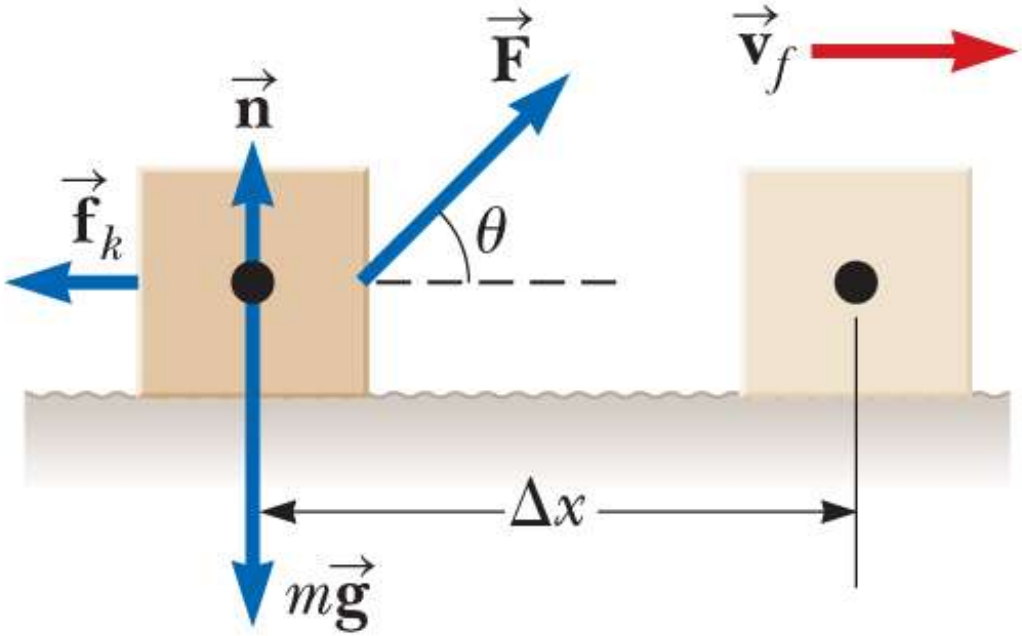
$$\left(\frac{1}{2}mv_f^{\,2} - 0\right) + \mu_k mgd = Fd$$

$$v_f = \sqrt{\frac{2}{m}\left(-f_k d + F\Delta x\right)}$$

$$v_f = \sqrt{2(3.0\text{ m})\left[\frac{12\text{ N}}{6.0\text{ kg}} - (0.15)(9.80\text{ m/s}^2)\right]} = \boxed{1.8\text{ m/s}}$$

Example 8.5:
A Block Pulled on a Rough Surface

(B) Suppose the force $\vec{\mathbf{F}}$ is applied at an angle θ as shown in the figure. At what angle should the force be applied to achieve the largest possible speed after the block has moved 3.0 m to the right?



Example 8.5:
A Block Pulled on a Rough Surface

$$\Delta K + \Delta E_{\text{int}} = W \qquad W = F\Delta x\cos\theta = Fd\cos\theta$$



$$\sum F_y = n + F\sin\theta - mg = 0$$

$$n = mg - F\sin\theta$$

$$\left(K_f - 0\right) + f_k d \rightarrow K_f = W_f - f_k d$$

$$K_f = Fd\cos\theta - \mu_k nd = Fd\cos\theta - \mu_k\left(mg - F\sin\theta\right)d$$

$$\frac{dK_f(\theta)}{d\theta} = -Fd\sin\theta - \mu_k\left(0 - F\cos\theta\right)d = 0$$

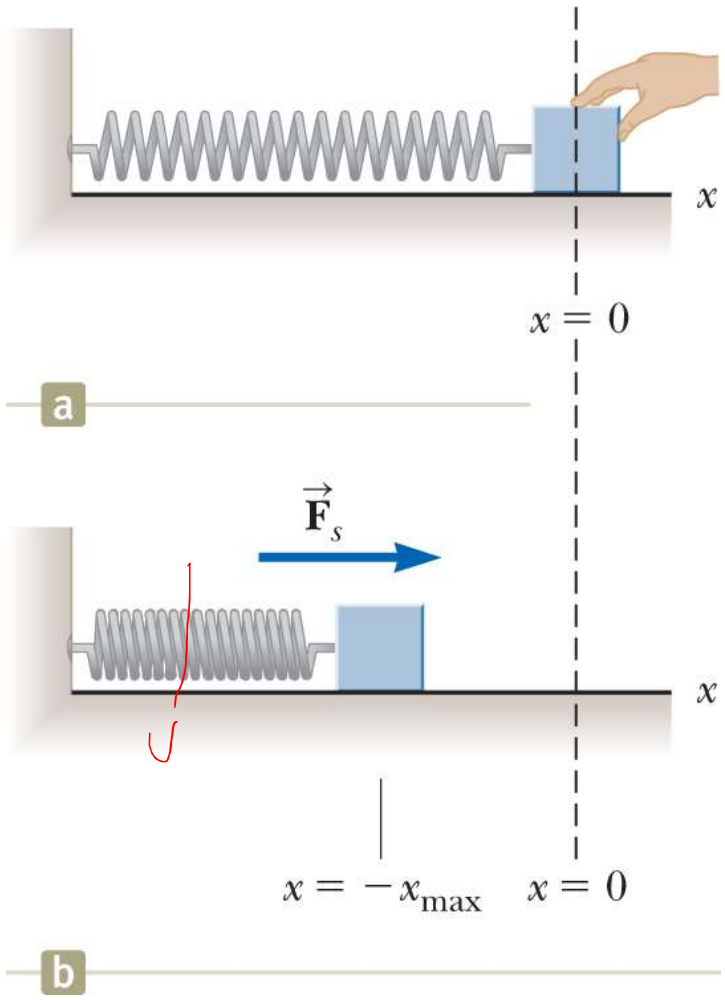
$$-\sin\theta + \mu_k\cos\theta = 0 \Rightarrow \tan\theta = \mu_k$$

$$\theta = \tan^{-1}\left(\mu_k\right) = \tan^{-1}\left(0.15\right) = \boxed{8.5^\circ}$$

Example 8.6:
A Block-Spring System

A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 1 000 N/m as shown in the top figure. The spring is compressed 2.0 cm and is then released from rest as in the bottom figure.

(A) Calculate the speed of the block as it passes through the equilibrium position $x = 0$ if the surface is frictionless.



Example 8.6:
A Block-Spring System

$v_i = 0$ at $x_i = -2.0$ cm, what is v_f at $x_f = 0$?

$$\Delta K + \Delta U = W_s \qquad W_s = \frac{1}{2}kx_{\max}^2$$

$$\frac{1}{2}mv_f^2 - 0 = \frac{1}{2}kx_{\max}^2 \Rightarrow v_f = x_{\max}\sqrt{\frac{k}{m}}$$

$$v_f = (0.020\ \text{m})\sqrt{\frac{1000\ \text{N/m}}{1.6\ \text{kg}}} = \boxed{0.50\ \text{m/s}}$$

Example 8.6: A Block-Spring System

(B) Calculate the speed of the block as it passes through the equilibrium position if a constant friction force of 4.0 N retards its motion from the moment it is released.

$$\Delta U + \Delta K + \Delta E_{\text{int}} = W_s \quad \Rightarrow \quad \left(\frac{1}{2}mv_f^2 - 0\right) + f_k d = W_s = \frac{1}{2}kx_{\max}^2$$

$$v_f = \sqrt{\frac{2}{m}\left(W_s - f_k d\right)} \qquad v_f = \sqrt{\frac{2}{m}\left(\frac{1}{2}kx_{\max}^2 - f_k d\right)}$$

$$v_f = \sqrt{\frac{2}{1.6\ \text{kg}}\left[\frac{1}{2}(1000\ \text{N/m})(0.20\ \text{m})^2 - (4.0\ \text{N})(0.020\ \text{m})\right]}$$

$$= 0.39\ \text{m/s}$$

Example 8.6:
A Block-Spring System

What if the friction force were increased to 10.0 N?
What is the block's speed at $x = 0$?

$$f_k d = (10.0\ \text{N})(0.20\ \text{m}) = 0.20\ \text{J}$$

$V_f = 0$

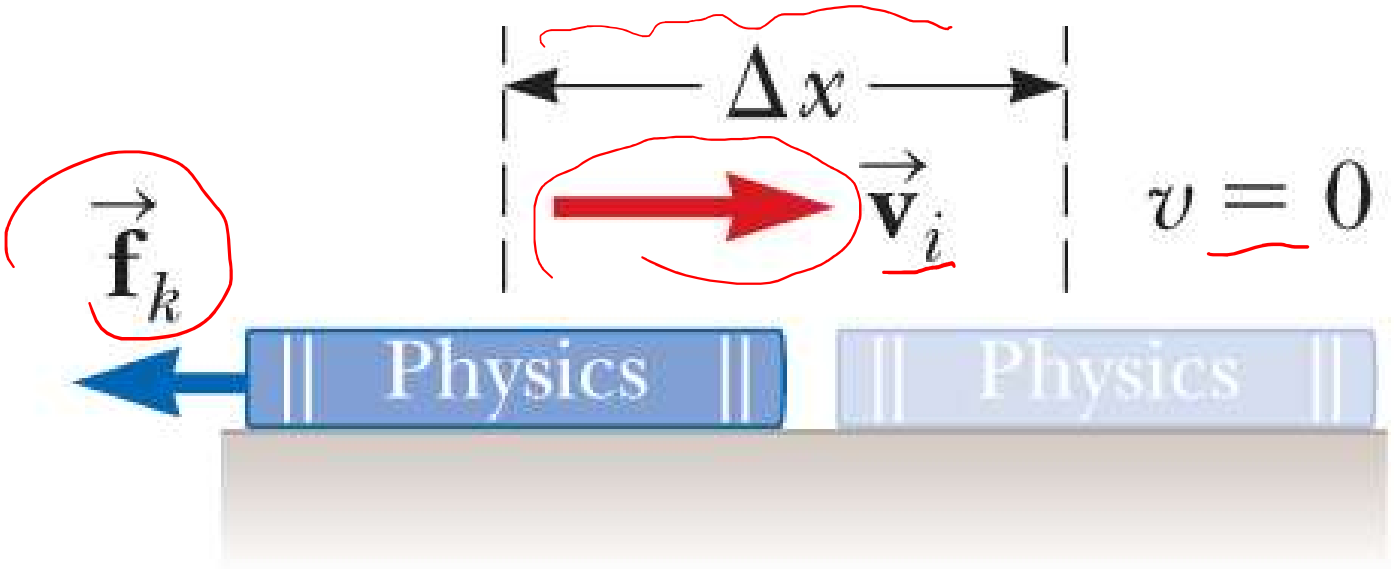
$\Delta KE = \frac{1}{2} m V_f^2 = \frac{1}{2} 4.6 \cdot (0.5)^2 = 0.8 \cdot (0.25)$

$= \frac{258}{1000} = \frac{200}{1000} = 0.2\ \text{J}$

Changes in Mechanical Energy for Nonconservative Forces

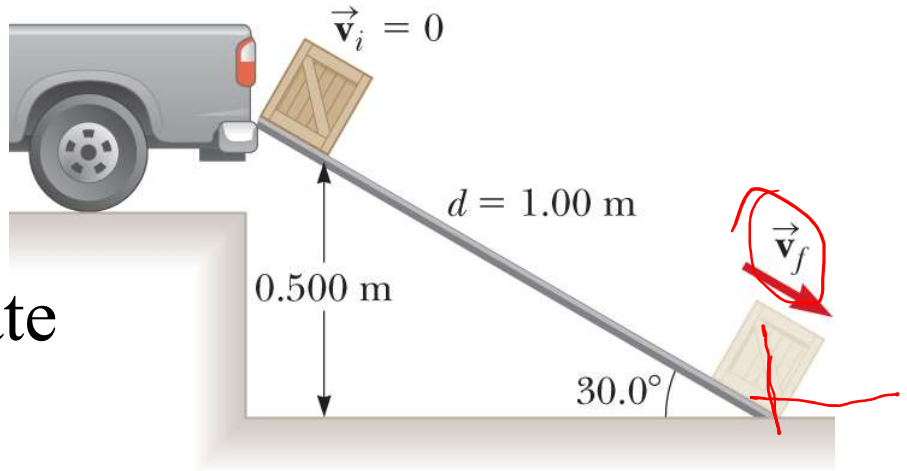
$$\Delta K + \Delta U_g + \Delta E_{\text{int}} = 0$$

non-conservative energy goes there



Example 8.7:
Crate Sliding Down a Ramp

A 3.00-kg crate slides down a ramp. The ramp is 1.00 m in length and inclined at an angle of 30.0° as shown in the figure. The crate starts from rest at the top and experiences a constant friction force of magnitude 5.00 N. The crate continues to move a short distance on the horizontal floor after it leaves the ramp, and then comes to rest.



(A) Use energy methods to determine the speed of the crate at the bottom of the ramp.

Example 8.7:
Crate Sliding Down a Ramp

Earth-crate system isolated

$$\Delta K + \Delta U_g + \Delta E_{\text{int}} = 0$$

$$\left(\frac{1}{2}mv_f^{\,2} - 0\right) + \left(0 - mgy_i\right) + f_k d = 0$$

$$v_f = \sqrt{\frac{2}{m}\left(mgy_i - f_k d\right)}$$

$$v_f = \sqrt{\frac{2}{3.00\ \text{kg}}\left[(3.00\ \text{kg})(9.80\ \text{m/s}^2)(0.500\ \text{m}) - (5.00\ \text{N})(1.00\ \text{m})\right]}$$

$$= \boxed{2.54\ \text{m/s}}$$

✓

Using f_k

Example 8.7:
Crate Sliding Down a Ramp

(B) How far does the crate slide on the horizontal floor if it continues to experience a friction force of magnitude 5.00 N?



$$\Delta K + \Delta E + \Delta E_{\text{int}} = 0$$

$$(0 - 0) + (0 - mgy_i) + f_k d_{\text{total}} = 0$$

$$d_{\text{total}} = \frac{mgy_i}{f_k} = \frac{(3.00\ \text{kg})(9.80\ \text{m/s}^2)(0.500\ \text{m})}{5.00\ \text{N}} = \boxed{2.94\ \text{m}}$$

Example 8.7:
Crate Sliding Down a Ramp

A cautious worker decides that the speed of the crate when it arrives at the bottom of the ramp may be so large that its contents may be damaged. Therefore, he replaces the ramp with a longer one such that the new ramp makes an angle of 25.0° with the ground. Does this new ramp reduce the speed of the crate as it reaches the ground?

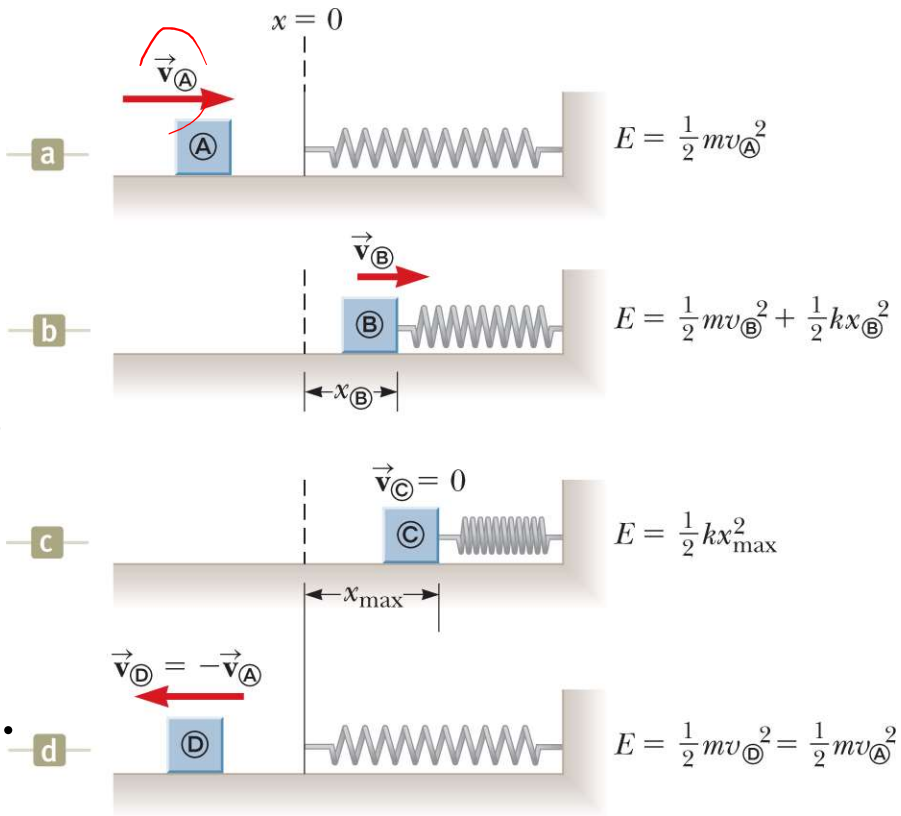
$$\sin 25.0^\circ = \frac{0.500\ \text{m}}{d} \rightarrow d = \frac{0.500\ \text{m}}{\sin 25.0^\circ} = 1.18\ \text{m}$$

$$v_f = \sqrt{\frac{2}{3.00\ \text{kg}}\left[(3.00\ \text{kg})(9.80\ \text{m/s}^2)(0.500\ \text{m}) - (5.00\ \text{N})(1.18\ \text{m})\right]}$$

$$= \boxed{2.42\ \text{m/s}}$$

Example 8.8:
Block-Spring Collision

A block having a mass of 0.80 kg is given an initial velocity $v_A = 1.2$ m/s to the right and collides with a spring whose mass is negligible and whose force constant is $k = 50$ N/m as shown in the figure.



(A) Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision.

Example 8.8:
Block-Spring Collision

$$\Delta K + \Delta U = 0$$

$$\left(0 - \frac{1}{2} m v_{\mathrm{A}}^{\ 2}\right) + \left(\frac{1}{2} k x_{\max}^{2} - 0\right) = 0$$

$$x_{\max} = \sqrt{\frac{m}{k}} v_{\mathrm{A}} = \sqrt{\frac{0.80\ \mathrm{kg}}{50\ \mathrm{N/m}}} (1.2\ \mathrm{m/s}) = \boxed{0.15\ \mathrm{m}}$$

Example 8.8:
Block-Spring Collision

(B) Suppose a constant force of kinetic friction acts between the block and the surface, with $\mu_k = 0.50$. If the speed of the block at the moment it collides with the spring is $v_A = 1.2$ m/s, what is the maximum compression x_C in the spring?

Example 8.8: Block-Spring Collision

$$f_k = \mu_k n = \mu_k mg$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\left(0 - \frac{1}{2} m v_{\text{A}}^{\ 2}\right) + \left(\frac{1}{2} k x_{\text{C}}^{\ 2} - 0\right) + \mu_k mg x_{\text{C}} = 0$$

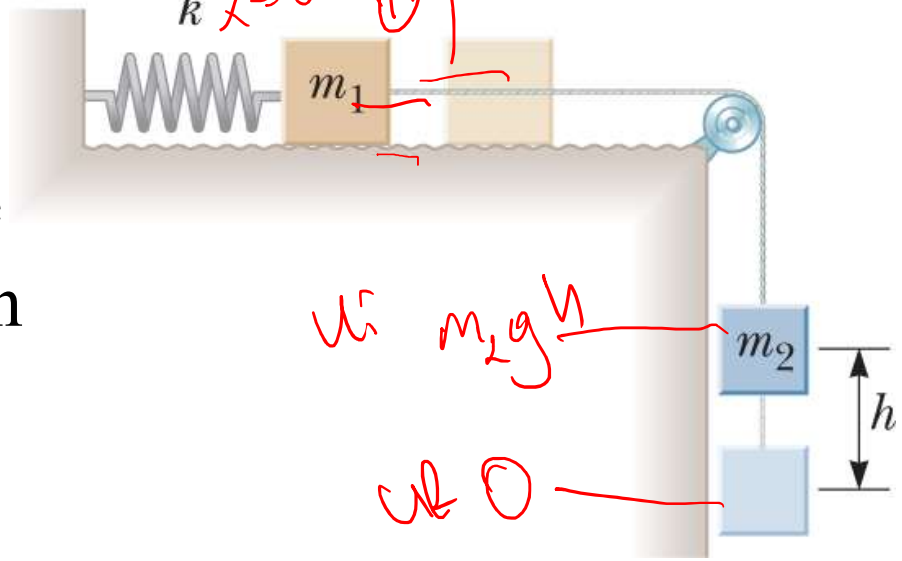
$$k x_{\text{C}}^{\ 2} + 2\mu_k mg x_{\text{C}} - m v_{\text{A}}^{\ 2} = 0$$

$$x_{\text{C}} = \frac{\mu_k mg}{k}\left(\pm\sqrt{1 + \frac{k v_{\text{A}}^2}{\mu_k^{\ 2} m g^2}} - 1\right)$$

$$x_{\text{C}} = \boxed{0.092 \text{ m}} \text{ or } -0.25 \text{ m}$$

Example 8.9:
Connected Blocks in Motion

Two blocks are connected by a light string that passes over a frictionless pulley as shown in the figure. The block of mass m_1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface.



Example 8.9: Connected Blocks in Motion

$$\Delta U_g = U_{gf} - U_{gi} \qquad \Delta U_s = U_{sf} - U_{si}$$

$$\Delta K + \Delta U_g + \Delta U_s + \Delta E_{\text{int}} = 0$$

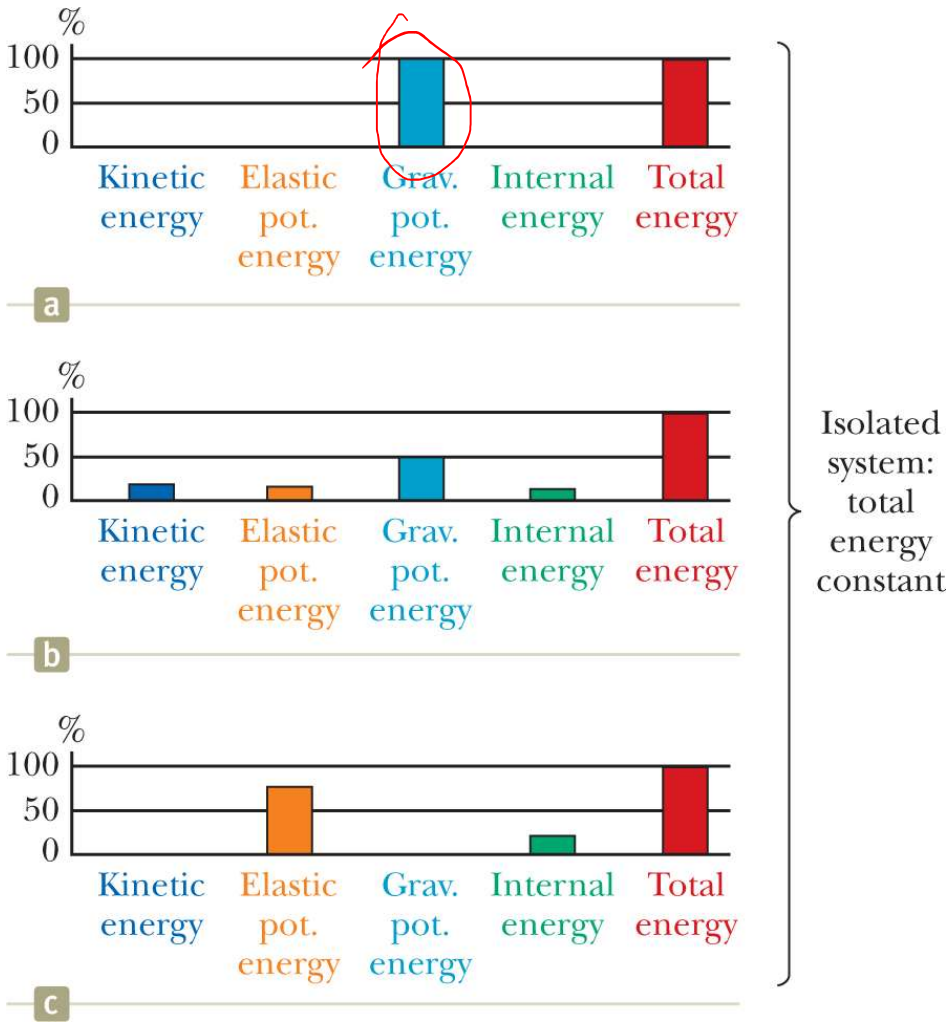
$$(0-0) + (0 - m_2 g h) + \left(\frac{1}{2}kh^2 - 0\right) + f_k h = 0$$

$$-m_2 g h + \frac{1}{2}kh^2 + \mu_k m_1 g h = 0$$

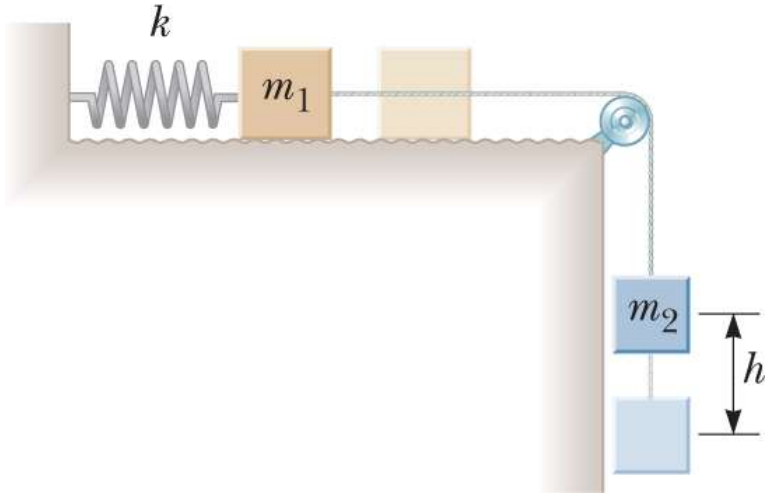
$$\mu_k = \frac{m_2 g - \frac{1}{2}kh}{m_1 g}$$

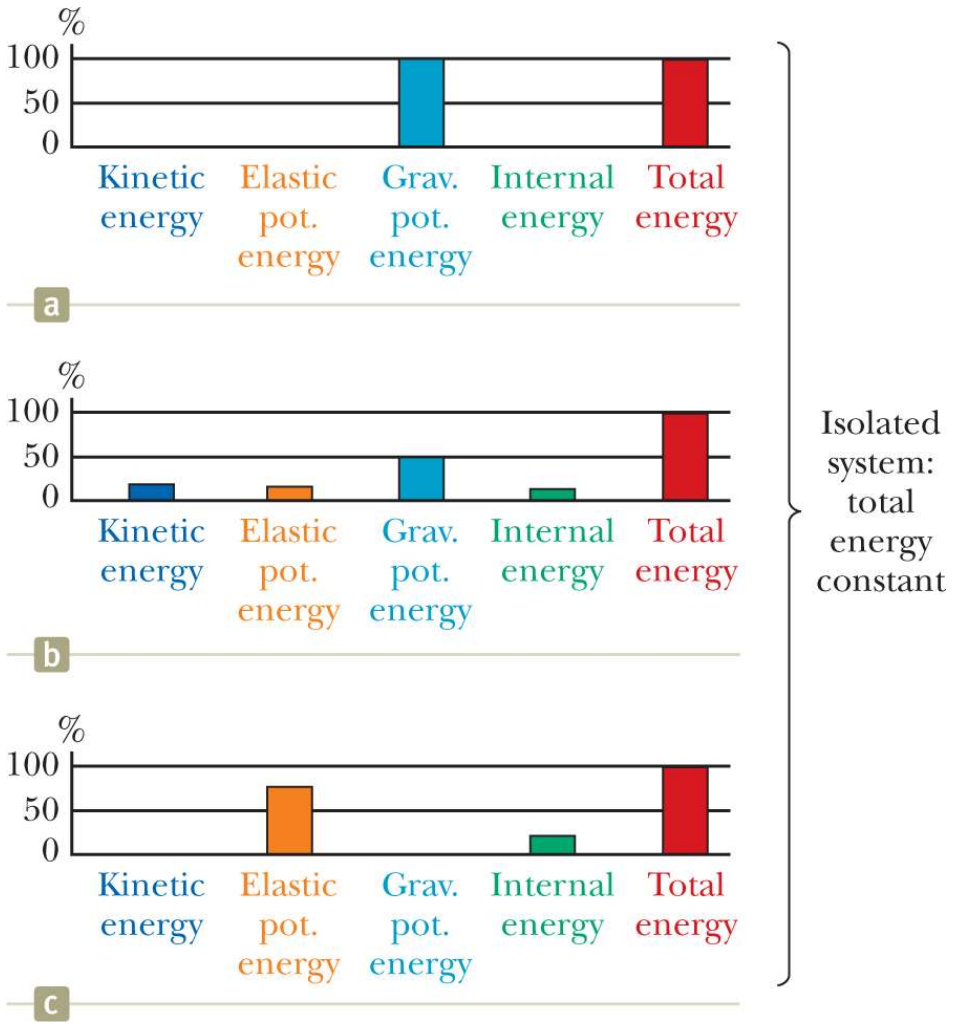
Conceptual Example 8.10: Interpreting the Energy Bars

The energy bar charts in the figure show three instants in the motion of the system described in the previous example. For each bar chart, identify the configuration of the system that corresponds to the chart.



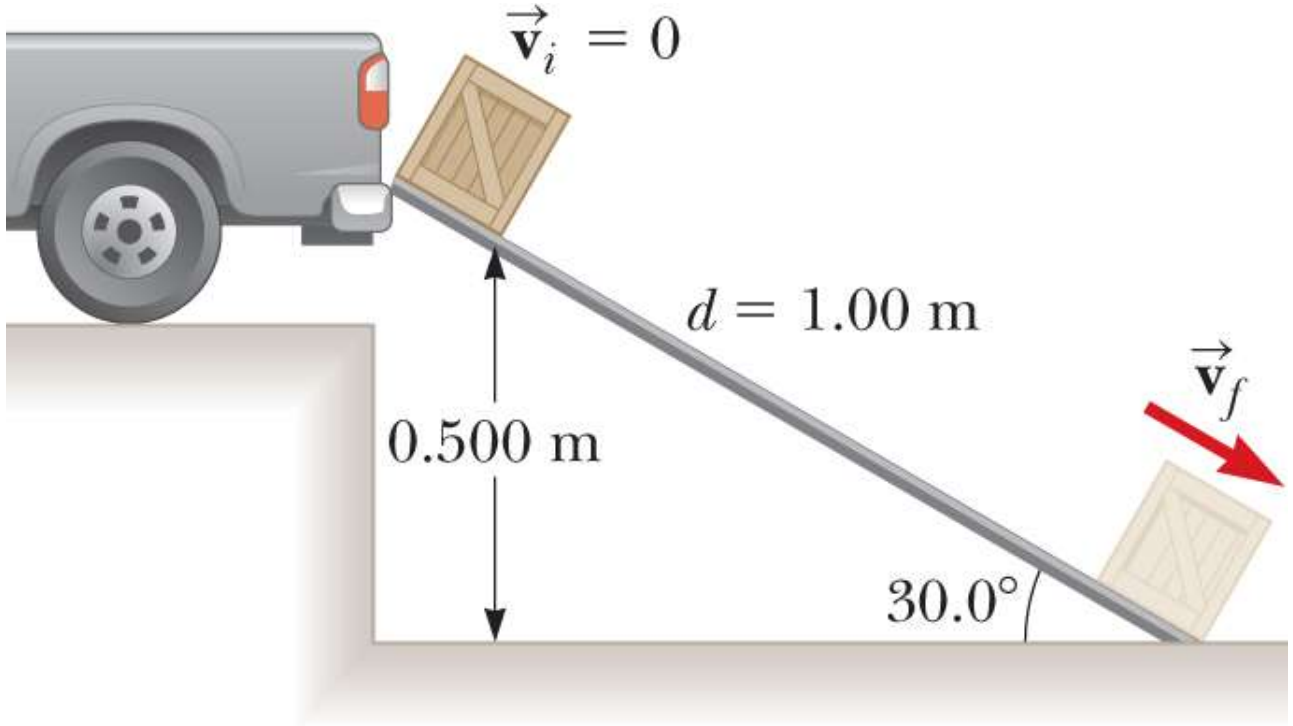
Conceptual Example 8.10: Interpreting the Energy Bars





Power

$$P \equiv \frac{dE}{dt} \qquad P_{\text{avg}} = \frac{W}{\Delta t}$$



Instantaneous Power

$$P = \lim_{\Delta t \to 0} \frac{W}{\Delta t} = \frac{dW}{dt}$$

$$dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

$$P = \frac{dW}{dt} = \vec{\mathbf{F}} \cdot \frac{d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}$$

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt}$$

Units of Power

$$1\ \text{W} = 1\ \text{J/s} = 1\ \text{kg}\cdot\text{m}^2/\text{s}^3$$

$$1\ \text{hp} = 746\ \text{W}$$

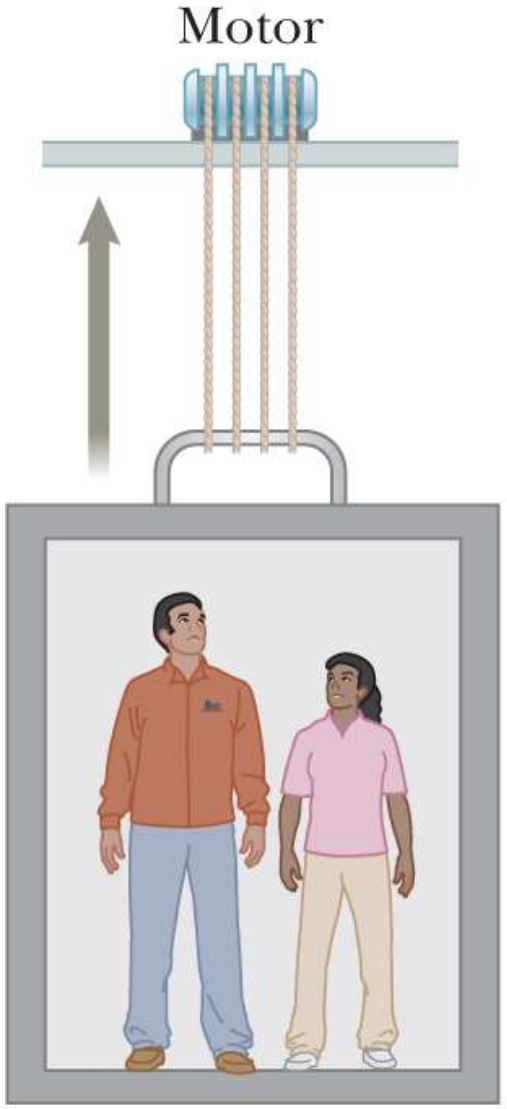
$$1\ \text{kWh} = \left(10^3\ \text{W}\right)\left(3\,600\ \text{s}\right) = 3.60\times10^6\ \text{J}$$

$$\left(0.100\ \text{kW}\right)\left(1.00\ \text{h}\right) = 0.100\ \text{kWh} = 3.60\times10^5\ \text{J}$$

Example 8.11:
Power Delivered by an Elevator Motor

An elevator car has a mass of 1 600 kg and is carrying passengers having a combined mass of 200 kg. A constant friction force of 4 000 N retards its motion.

(A) How much power must a motor deliver to lift the elevator car and its passengers at a constant speed of 3.00 m/s?

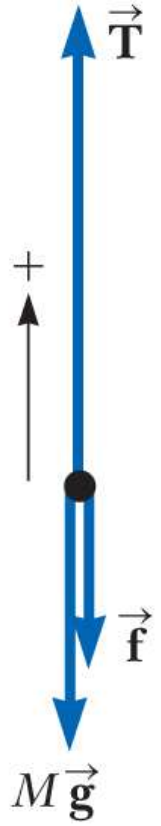


Example 8.11:
Power Delivered by an Elevator Motor

$$\sum F_y = T - f - Mg = 0$$

$$T = Mg + f$$

$$P = \vec{\mathbf{T}} \cdot \vec{\mathbf{v}} = Tv = \left(Mg + f\right)v$$



$$P = \left[\left(1800\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right) + \left(4000\ \text{N}\right)\right]\left(3.00\ \text{m/s}\right)$$

$$= 6.49 \times 10^4\ \text{W}$$

Example 8.11:
Power Delivered by an Elevator Motor

(B) What power must the motor deliver at the instant the speed of the elevator is v if the motor is designed to provide the elevator car with an upward acceleration of 1.00 m/s^2?

Example 8.11:
Power Delivered by an Elevator Motor

$$\sum F_y = T - f - Mg = Ma$$

$$T = M\left(a + g\right) + f$$

$$P = Tv = \left[M\left(a + g\right) + f\right]v$$

$$P = \left[\left(1800\ \text{kg}\right)\left(1.00\ \text{m/s}^2 + 9.80\ \text{m/s}^2\right) + 4000\ \text{N}\right]v$$

$$= \boxed{\left(2.34 \times 10^4\ \text{N}\right)v}$$

$$P = \left(2.34 \times 10^4\right)\left(3.00\ \text{m/s}\right) = 7.02 \times 10^4\ \text{W}$$